

PRATEEK KUMAR PANDA

+91-9827703357 | paandaprateek0201@gmail.com | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

Results-driven Data Scientist and AI/ML Engineer with 2.5+ years of experience building production-grade machine learning systems, generative AI platforms, and advanced analytics solutions. Proven expertise in designing multi-agent LLM frameworks, NLP pipelines, and large-scale data processing with PySpark and cloud platforms (AWS, Azure). Adept at translating complex business problems in pharma and QSR industries into data-driven models that drive measurable revenue impact and operational efficiency.

TECHNICAL SKILLS

ML / AI: Machine Learning, Deep Learning, NLP, Generative AI, RAG (Retrieval-Augmented Generation), RLHF, XGBoost, Random Forest, Clustering (K-Means, Hierarchical), Boruta Feature Selection, Sentiment Analysis, Price Elasticity Modeling, Sales Decomposition, Cannibalization & Halo Analysis

LLMs & Agents: LangChain, CrewAI, Ollama, AWS Bedrock (Claude Sonnet 3.5), OpenAI Whisper (STT), ElevenLabs (TTS), Pinecone (Vector DB), MCP (Model Context Protocol)

Big Data & Analytics: PySpark, Databricks, Amazon Redshift, AWS Wrangler, DuckDB, DBeaver, Power BI, Streamlit

Cloud & Databases: AWS (Bedrock, S3, EC2, CloudWatch), Azure Blob Storage, ADLS, MongoDB, MySQL

Backend & Programming: Python, FastAPI, Django, Flask, WebSockets, SQL, JavaScript, HTML/CSS

Tools & Platforms: Git, GitHub, Postman, Jupyter Notebook, Google Colab, Figma, Selenium, VSCode

PROFESSIONAL EXPERIENCE

Data Scientist | Involead Services Pvt. Ltd., On-site | Jan 2024 – Present

European Pharma – Production Multi-Agent AI Platform

- Architected a production-grade multi-agent AI system using CrewAI with Text, Speech, and Web Search agents, replacing a single-model setup with an orchestrated multi-modal pipeline.
- Integrated Google SERP via MCP for real-time search and leveraged AWS Bedrock (Claude Sonnet 3.5) with LangChain for LLM inference and pipeline orchestration.
- Built a low-latency speech pipeline using OpenAI Whisper (STT) and ElevenLabs (TTS) over WebSocket connections, enabling real-time conversational AI with near-zero turn latency.
- Implemented RAG pipeline (OCR → embedding → Pinecone) with metadata caching and S3 + DuckDB integration, significantly reducing query response time and infrastructure overhead.
- Served all inference endpoints via FastAPI with MongoDB for session persistence; delivered to production, improving client-facing AI interaction quality.

European Pharma – HCP Profiling & Content Optimisation

- Built Healthcare Professional (HCP) profiles using Email CTR, web traffic data, and unsupervised clustering models (K-Means, Hierarchical) to assess content effectiveness and audience segmentation.
- Developed a Streamlit-based simulation tool enabling clients to tune KPIs via content metadata, supporting data-driven targeting and eliminating manual trial-and-error.
- Applied XGBoost for KPI simulation and Boruta for feature/metadata optimisation, replacing ad-hoc content strategy with a systematic, repeatable analytical framework.
- Ingested and queried large-scale healthcare data from Amazon Redshift using AWS Wrangler and DBeaver.

USA QSR – Transactional Analytics & Revenue Intelligence

- Ingested and processed large-scale transactional data from Azure Blob Storage via Databricks (PySpark); delivered revenue insights that informed Value Menu restructuring and item-level pricing strategy.
- Modelled price elasticity, sales decomposition, demand transfer, and cannibalization & halo effects to optimise item mix and reduce revenue leakage.
- Assessed new menu item launch impact using historical trial rates, improving forecast accuracy for menu expansion planning.
- Automated weekly data refresh using cron jobs aligned to a 5-4-4 fiscal calendar; surfaced actionable insights through Power BI dashboards connected to ADLS, reducing manual reporting effort.

EDUCATION

B.Tech in Computer Science & Technology | GIET University, Odisha | 2020 – 2024

CGPA: 8.29 / 10.0

CERTIFICATIONS

- Oracle Cloud Infrastructure – Generative AI Certified Professional

KEY ACHIEVEMENTS & IMPACT

- Delivered a production multi-agent AI platform (CrewAI) for a European pharma client, improving AI interaction quality and enabling real-time conversational workflows.
- Informed Value Menu restructuring and item-level pricing for a major US QSR chain through price elasticity, cannibalization, and halo impact modelling.
- Automated weekly data pipelines and Power BI dashboards, significantly reducing manual reporting effort and freeing bandwidth for strategic analysis.
- Replaced ad-hoc pharma content strategies with a systematic XGBoost + Boruta ML framework, driving measurable improvements in HCP targeting accuracy.